

AI-Powered Customer Churn & Revenue Analytics

Data Analytics + Artificial Intelligence Project Report

1. Abstract

This project combines descriptive data analytics with an artificial intelligence component to analyze customer behavior and predict churn risk. A synthetic customer dataset is cleaned, explored and transformed into business KPIs. A Logistic Regression model then predicts customer churn probability, classifies customers into risk levels and identifies the strongest model signals. The pipeline also demonstrates SQL and automated visualization.

2. Problem Statement

Businesses need to understand not only what happened in their customer base but also which customers may be at risk in the future. Traditional reporting can describe churn after it happens. This project adds a predictive layer so analysts can prioritize customers for retention analysis.

3. Objectives

- Perform data cleaning and exploratory analytics.
- Calculate customer and revenue-related KPIs.
- Train an AI/ML model to predict churn.
- Generate customer-level churn probabilities.
- Identify important churn drivers.
- Estimate annual customer value associated with high-risk customers.
- Demonstrate SQL-based business analysis.
- Generate reusable charts and reports.

4. Dataset

The program generates 6,000 synthetic customer records. Variables include tenure, monthly charges, support tickets, late payments, usage hours, contract, payment method, region, plan, support plan and churn status. A small number of missing values are deliberately introduced so that the cleaning stage can be demonstrated.

5. Data Preparation

The cleaning pipeline removes duplicate customer IDs, converts numeric columns, fills missing numeric values with medians and categorical values with the mode, and engineers annual customer value from monthly charges.

6. Descriptive Analytics

The project calculates total customers, churned customers, churn rate, average monthly charges and estimated annual customer value associated with churned customers. It also compares churn rates across contract types.

7. AI Model

A Logistic Regression classifier is used because it is suitable for binary classification and provides interpretable coefficients. Numeric variables are standardized, while categorical variables are one-hot encoded. The dataset is divided into training and test sets using stratified sampling.

8. Model Evaluation

The model is evaluated using accuracy, precision, recall, F1 score and ROC-AUC. These metrics provide different perspectives on predictive performance, especially the ability to identify churn cases.

9. Explainable AI

The project extracts model coefficients and ranks features by absolute impact. Positive coefficients indicate an association with higher predicted churn probability, while negative coefficients indicate an association with lower predicted churn probability, within this fitted model.

10. Customer Risk Scoring

Every customer receives a churn probability. Customers are grouped into Low, Medium and High risk using predefined probability ranges. A recommended analytical action is also generated for each risk group.

11. SQL Analysis

The cleaned customer data is stored in SQLite. A SQL query calculates customer count, average monthly charges and churn rate by contract type. This demonstrates that the project can combine Python analytics with relational database querying.

12. Business Applications

The workflow can support retention analysis, customer-service prioritization, marketing segmentation and management reporting. In a real company, the synthetic generator would be replaced with governed customer data from a warehouse, CRM or operational database.

13. Responsible AI Considerations

The dataset in this project is synthetic. A real deployment would require validation for data quality, model drift, fairness, privacy and appropriate human review. Predictions should support analysis rather than automatically determine customer treatment.

14. Conclusion

This project demonstrates a complete Data Analytics + AI workflow: data generation, cleaning, KPI analysis, SQL, predictive modeling, explainability, customer risk scoring and automated reporting. It provides a portfolio-ready example of how an analyst can move from descriptive analytics to predictive insights.

Appendix: Submission Structure

File	Purpose
ai_customer_churn_analytics.py	Main analytics and AI/ML pipeline
requirements.txt	Python dependencies
README.md	Setup and project documentation
Project_Report.pdf	Submission report
output/	Generated datasets, charts, database and AI insights